Description

Image





Caption

1. Strands of copper electrical wire. 2. A dome on the newly reconstructed Berliner Schloss, Germany. The copper cladding has not yet developed its characteristic verdigris. 3. Cladding around a window in a tower of the Knippelsbro, Copenhagen, Denmark. The uneven green patina is typical of copper oxidation. Images used under license from Shutterstock.com

The material

In Victorian times you washed your clothes in a 'copper' - a vat or tank of beaten copper sheet, heated over a fire; the device exploited both the high ductility and the thermal conductivity of the material. Copper has a distinguished place in the history of civilization: it enabled the technology of the Bronze age (3000 BC - 1000 BC). It is used in many forms: as pure copper, as copper-zinc alloys (brasses), as copper-tin alloys (bronzes), and as copper-nickel and copper-beryllium. The designation of 'copper' is used when the percentage of copper is more than 99.3%. It is heavy and soft, not what you want for good mechanical performance. It is in its thermal, electrical, environmental and aesthetic qualities that copper excels. Its thermal and electrical conductivities are greater than those of any other metal except silver, and it is 50 times more expensive. It is exceptionally durable -- copper artifacts 2500 years old survive today; it is one of the reasons that it is used for coinage. And it is visually attractive: copper objects are prized, and prestige buildings are clad with it.

Compositional summary

Cu

General properties

Density	8.94e3			kg/m^3
Price	* 6.58	-	6.75	GBP/kg
Date first used	-8000			

Mechanical properties

Young's modulus	120	-	140	GPa

Shear modulus		44	-	50	GPa
Bulk modulus	*	130	-	150	GPa
Poisson's ratio		0.34	-	0.35	
Yield strength (elastic limit)		50	-	340	MPa
Tensile strength		211	-	380	MPa
Compressive strength	*	50	-	340	MPa
Elongation		6	-	50	% strain
Hardness - Vickers		46.3	-	115	HV
Fatigue strength at 10^7 cycles		70.6	-	115	MPa
Fracture toughness	*	38.8	-	95	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	9.3e-5	-	0.0013	

Thermal properties

Melting point		1.06e3	-	1.09e3	°C
Maximum service temperature		76.9	-	120	°C
Minimum service temperature		-273			°C
Thermal conductor or insulator?		Good conductor			
Thermal conductivity		186	-	398	W/m.°C
Specific heat capacity		378	-	387	J/kg.°C
Thermal expansion coefficient		16.8	-	17.9	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good conductor			
Electrical resistivity		1.7	-	4	µohm.cm

Optical properties

Transparency		Opaque			

Critical Materials Risk

High critical material risk?		Yes			

Processability

Castability		3	-	5	
Formability		4	-	5	
Machinability		4	-	5	
Weldability		3			
Solder/brazability		5			

Durability: water and aqueous solutions

Water (fresh)		Excellent			

Water (salt)	Excellent
Soils, acidic (peat)	Acceptable
Soils, alkaline (clay)	Excellent
Wine	Excellent

Durability: acids

Acetic acid (10%)	Limited use
Acetic acid (glacial)	Unacceptable
Citric acid (10%)	Excellent
Hydrochloric acid (10%)	Acceptable
Hydrochloric acid (36%)	Unacceptable
Hydrofluoric acid (40%)	Limited use
Nitric acid (10%)	Acceptable
Nitric acid (70%)	Unacceptable
Phosphoric acid (10%)	Limited use
Phosphoric acid (85%)	Unacceptable
Sulfuric acid (10%)	Unacceptable
Sulfuric acid (70%)	Unacceptable

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Excellent

Durability: fuels, oils and solvents

Amyl acetate	Excellent
Benzene	Excellent
Carbon tetrachloride	Excellent
Chloroform	Excellent
Crude oil	Acceptable
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Limited use
Toluene	Excellent
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Limited use
Acetone	Excellent
Ethyl alcohol (ethanol)	Excellent
Ethylene glycol	Excellent
Formaldehyde (40%)	Excellent
Glycerol	Excellent
Methyl alcohol (methanol)	Excellent

Durability: halogens and gases

Chlorine gas (dry)	Excellent
Fluorine (gas)	Acceptable
O2 (oxygen gas)	Limited use
Sulfur dioxide (gas)	Limited use

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Excellent

Durability: flammability

Flammability	Non-flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Excellent
Tolerance up to 150 C (302 F)	Excellent
Tolerance up to 250 C (482 F)	Unacceptable
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	1.85e7	tonne/yr
Reserves, principal component	6.9e8	tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 4.83	-	5.32	kg/kg
Embodied energy (virgin grade production)	* 72.3	-	79.7	MJ/kg
Water usage	* 293	-	324	l/kg

Material processing: energy

Metal casting, embodied energy	* 41.5	-	46.2	MJ/kg
Metal extrusion, hot, embodied energy	* 9.56	-	17.2	MJ/kg
Metal rolling and forging, embodied energy	* 5.25	-	9.17	MJ/kg
Wire drawing, embodied energy	* 13.1	-	14.5	MJ/kg
Metal powder forming, embodied energy	* 41.8	-	46.8	MJ/kg
Chemical vapor deposition (CVD), embodied energy	* 7.41e3	-	8.52e3	MJ/kg
Machining, coarse, embodied energy	* 15.9	-	18.5	MJ/kg
Machining, fine, embodied energy	* 18.5	-	23.4	MJ/kg
Machining, grinding, embodied energy	* 21.1	-	28.3	MJ/kg
Machining, laser, CO2, embodied energy	* 0.882	-	1.01	MJ/m

Material processing: climate change

Metal casting, climate change (CO2-eq)	* 2.92	-	3.26	kg/kg
Metal extrusion, hot, climate change (CO2-eq)	* 0.626	-	1.13	kg/kg
Metal rolling and forging, climate change (CO2-eq)	* 0.344	-	0.601	kg/kg
Wire drawing, climate change (CO2-eq)	* 0.86	-	0.95	kg/kg
Metal powder forming, climate change (CO2-eq)	* 3.39	-	3.79	kg/kg
Chemical vapor deposition (CVD), climate change (CO2-eq)	* 610	-	701	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.04	-	1.22	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.21	-	1.54	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.38	-	1.86	kg/kg
Machining, laser, CO2, climate change (CO2-eq)	* 0.0726	-	0.0834	kg/m

Recycling and end of life

Recycle	✓			
Functional recycle	✓			
Climate change (CO2-eq), recycling	* 1.06	-	1.17	kg/kg
Embodied energy, recycling	* 17.4	-	19.1	MJ/kg
Recycle fraction in current supply	28			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

Copper and its alloys are particularly easy to recycle - in many countries the recycle fraction approaches 90%.

Supporting information

Design guidelines

Copper and its alloys are easy to cast, to roll to sheet, to draw to wire, and to shape in other ways. They resist corrosion in the atmosphere, acquiring an attractive green patina (copper carbonate) in clean air, and a black one (copper sulfide) in one that is not - copper roofs in cities are usually black. The patina of bronze is a rich warm brown, much loved by sculptors. Pure copper has excellent electrical and thermal conductivity, is easy to fabricate and join, has good corrosion resistance and reasonable strength. Where high conductivity is necessary, oxygen-free high-conductivity (OFHC) copper is used. In its annealed form it is soft and ductile; with work-hardening the material becomes harder but less ductile.

Technical notes

There is now a UNS designation system for copper and its alloys: the letter C (for 'copper') followed by a 5-digit number. Only the first digit means anything: C1**** designates almost pure copper, the C2, C3 and C4 series are brasses with increasing zinc content, the C5s are bronzes based on copper and tin, the C6s are other bronzes containing aluminum instead of tin, and the C7s are copper-nickel alloys.

Typical uses

Electrical wiring, electrical conductors, cables, busbars, high strength, high conductivity wires and sections, overheads lines, contact wires, resistance-welding electrodes, terminals, high conductivity items for use at raised temperatures, heat exchangers, heat sinks, coinage, pans, kettles and boilers, plates for etching and engraving, roofing and architecture, boilers and pressure vessels.

Links

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Reference